

OpenRNG

Patent Portfolio



COMPUTATIONAL TRUST INFRASTRUCTURE

5-PATENT FAMILY

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OpenRNG · Taipei, Taiwan

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SECTION I

Executive Summary

Thesis

Computational trust is a new infrastructure layer. Every random number has a trust problem — traditional systems return a value and discard the process that created it. OpenRNG introduces **Verifiable Entropy** — transforming randomness from a disposable output into a portable, auditable, and independently verifiable digital object.

The Five Patents — Logical Curve

#	TECHNICAL PROBLEM	SYSTEM PROPERTY	STATUS	PATENTABILITY
1	Random values cannot be individually verified	Verifiable Randomness	✅ Approved	—
2	Random values can be previewed before commitment	Dual Temporal Integrity	📄 Ready to file	High
3	Trust disappears after randomness is consumed	Portable Trust Assertion	📄 Ready to file	High
4	Computational trust cannot survive delegation	End-to-End Verifiable Trust	🔬 Problem defined	70–80%
5	Trust quality silently degrades during operation	Continuous Trust Health	🔬 Problem defined	80–90%

The Logical Progression

#1 Can randomness be verified? → #2 Can randomness be hidden until it should exist? → #3 Can trust survive after randomness is consumed? → #4 Can trust survive delegation? → #5 Can trust survive time?

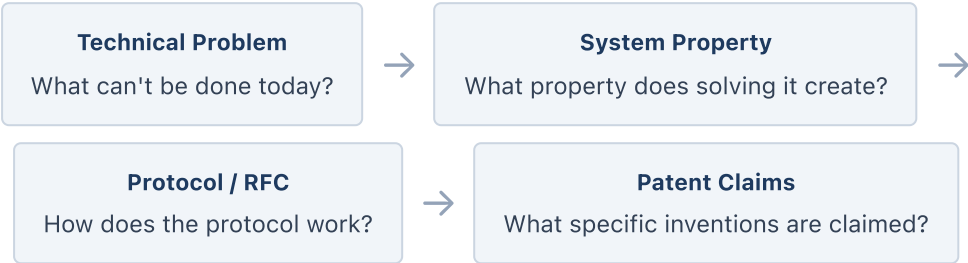
Strategic Value

- **Patents #1–#3:** Gaming / regulated industry foundation
- **Patent #4:** AI agent delegation (OpenAI, multi-agent systems)
- **Patent #5:** Highest commercial value — "Trust Health" as new category (like Battery Health)
- **Big Tech interest:** OpenAI → #4/#5 · Google/Microsoft → #5 (governance) · Gaming → #1–#3
- **Strategic decision:** Stop after #5 — next patent emerges from real usage

SECTION II

Framework

Problem → Property → Protocol → Patent



Dual Naming Convention

LAYER	PURPOSE	AUDIENCE
Technical Problem language	Precise problem definition	Patent examiners, engineers, attorneys
System Property language	Capability description	Papers, talks, standards, industry
Movement language	Vision & narrative	Manifestos, vision ("The Second Law of Computational Trust")

Rule: Never mix these layers.

PATENT #1

Verifiable Randomness

Technical Problem:	Random values cannot be independently verified	System Property:	Verifiable Randomness
Status:	 APPROVED	Title:	Method and System for Gaming Random Number Generation

The Problem

When a casino pays out \$2 million on a jackpot, regulators need to verify the winning number was genuinely random. Current systems produce audit logs written by the same system they audit. The randomness itself carries no proof.

The Invention

A system that generates random numbers as **Verifiable Entropy Objects (VEOs)** — cryptographic containers carrying entropy payload, source provenance, confidence metrics, provider signatures, and blockchain anchors. Any party can independently verify a VEO without trusting the provider.

Core Innovation

The **VEO-1 standard** — transforms randomness from a disposable value into a verifiable digital object with five layers: Payload, Provenance, Confidence, Proof, and Anchor.

Key Technical Components

- **VDF** (Verifiable Delay Functions) for time-locked entropy generation
- **drand** public randomness beacon for independent entropy sourcing
- **Merkle tree batching** for efficient verification of large token pools
- **Polygon blockchain anchoring** for immutable proof
- **Entropy Confidence Score (ECS):** 0–1000 quality metric across 6 dimensions

PATENT #2

Dual Temporal Integrity

Technical Problem: Random values can be previewed before commitment

System Property: Dual Temporal Integrity

Status:  **Ready to file**
(Provisional + Non-Provisional drafted)

The Problem

The "**Insider Preview Attack**" — in traditional RNG, the system generating randomness can pre-view values before they're committed. An operator, insider, or compromised system could select favorable outcomes.

The Invention

A dual-phase temporal commitment scheme where:

1. Random values are generated and cryptographically committed (Merkle root anchored on-chain) **BEFORE** any request arrives
2. Values are revealed only upon request, with the pre-commitment proving they existed before the request

Core Innovation

Separating the generation timeline from the consumption timeline through VDF-enforced time delays + blockchain anchoring, making preview attacks mathematically impossible.

Differentiation from Chainlink VRF

Chainlink uses request-response (randomness generated AFTER request). OpenRNG pre-generates pools of committed values — fundamentally different temporal model.

PATENT #3

Portable Trust Assertion

Technical Problem: Trust disappears after randomness is consumed

System Property: Portable Trust Assertion

Status:  **Ready to file**
(Provisional + Non-Provisional drafted)

The Problem

Once a random number is consumed, the trust evidence that backed it is discarded. A slot machine uses a random number, records the outcome, but the cryptographic proof of that number's integrity is lost. Trust is disposable.

The Invention

Trust Assertions that persist after entropy consumption — portable, verifiable records that any party can independently verify at any future time. The VEO becomes a permanent trust artifact, not a disposable value.

Core Innovation

The concept of "**entropy receipts**" — each consumption generates a portable attestation record that carries full provenance, quality scores, and verification paths. Trust becomes a first-class, persistent data object.

Key Continuation Anchors

This patent's non-provisional specification includes the Five-Layer Trust Stack and Decision Attestation Records — securing earliest priority dates for Patents #4 and #5.

PATENT #4

End-to-End Verifiable Trust

Technical Problem: Computational trust cannot survive delegation

System Property: End-to-End Verifiable Trust

Status: 🦋 Problem defined, drafts completed

Patentability: 70–80%

The Problem

When a trusted computation delegates work to another agent, the trust breaks. The delegation boundary destroys three things:

1. **Provenance linkage** — delegated agent gets task but not trust artifact
2. **Quality continuity** — no mechanism to detect trust quality degradation through delegation
3. **Audit continuity** — auditor cannot trace back to original trust assertion

Why Existing Approaches Fail

- **Task queues** (Celery, SQS) deliver tasks, not trust
- **API auth** (OAuth, JWT) proves WHO delegated, not WHY trustworthy
- **Distributed tracing** (OpenTelemetry) traces execution path, not trust path
- **Blockchain logs** record THAT delegation occurred, not the trust that AUTHORIZED it
- **PKI** proves identity delegation, not computational trust delegation

The Invention

A system for preserving computational trust across delegation boundaries by cryptographically linking child trust assertions to parent assertions through lineage records that enforce non-amplification, record degradation transparently, and enable independent verification of the complete trust chain from any node to root — without cooperation from intermediate nodes.

Target Market

Multi-agent AI systems (OpenAI, Anthropic), cross-organizational computation, regulatory compliance chains.

Action Needed: Add broader independent claim beyond DAG mechanics — let DAG be one embodiment, not the only claim.

PATENT #5

Continuous Trust Health

Technical Problem: Trust quality silently degrades during operation

System Property: Continuous Trust Health

Status:  **Problem defined, drafts completed**

Patentability: 80–90% (highest of all five)

The Problem — Trust Decay

Trust infrastructure doesn't fail catastrophically — it rots. Entropy sources silently degrade. Quality scores drift. Trust chains grow deeper than policy allows. Anchoring latency creeps up. Source diversity narrows. None triggers an error. Each is individually small. Collectively, they transform a AAA-grade system into LOW-grade — without anyone noticing.

The Battery Analogy

Trust Decay is to trust infrastructure what battery drain is to mobile devices. A fully charged battery (AAA trust) slowly loses charge. The drain is not an error — it is physics. Without monitoring, the user doesn't know until the device dies. Trust infrastructure needs the same: a trust health meter, decay detection, and active maintenance.

Why Existing Tools Fail

- **SIEM** (Splunk) monitors security events, not trust quality
- **APM** (Datadog) monitors performance — system is fast while trust silently degrades
- **GRC platforms** require manual periodic checks — Trust Decay is continuous
- **Alerting systems** catch threshold breaches — Trust Decay is sub-threshold (many small drifts)

The Invention

A system for detecting, quantifying, and counteracting continuous trust decay through:

- Multi-dimensional trust health monitoring
- Anomaly detection against trust-specific metrics (not security/performance)
- Automated corrective response (source rebalancing, policy tightening)
- Continuous machine-readable compliance attestation
- Predictive trust quality analytics forecasting decay before critical thresholds

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Commercial Value

Highest of all five patents. "Trust Health" creates a new monitoring category. Every compliance-heavy industry needs this. Google/Microsoft interested for governance/compliance.

SECTION III

Whitepaper: OpenRNG — The Verifiable Entropy Network

Core Concept

Verifiable Entropy Objects (VEO-1) — cryptographic containers for randomness carrying payload, provenance, confidence, proof, and anchor layers.

Object Classes

CLASS	NAME	DESCRIPTION
VEO-1A	Raw Source	Single-source entropy
VEO-1B	Composite	Multi-source aggregated
VEO-1C	Anchored	Composite + blockchain anchor
VEO-1D	Decision	Verifiable Decision Objects (reserved)

Entropy Confidence Score (ECS)

0–1000 quality metric measuring freshness, source diversity, independence, manipulation resistance, and verification success.

"Randomness is a value. Entropy is infrastructure." — Like electricity has meters, grids, and regulatory infrastructure, randomness needs the same.

SECTION IV

Prior Art Landscape

HIGH RISK**1. US12401532B2 — Chainlink (GRANTED)**

Distributed VRF with blockchain verification. Mentions VDF in disclosure.

Differentiation: Chainlink = distributed oracle + VRF (secret key). OpenRNG = centralized server + VDF (time-based) + Merkle batching. Different architecture, different primitive, different trust model.

2. US10291408B2 — Amazon (GRANTED)

Merkle tree generation from PRNG values.

Differentiation: Amazon = PoW mining challenges. OpenRNG = random token verification. Different use case entirely.

MEDIUM RISK**3. US20220410017A1 — nChain (ABANDONED)**

Provably fair gaming + blockchain.

Differentiation: nChain = interactive commit-reveal. OpenRNG = non-interactive pre-generated batches.

4. WO2017053754A1 — Spur Trail (CEASED)

Generic provably fair gaming. No Merkle batching, no VDF.

Key Differentiation

OpenRNG's sub-2ms in-memory token injection + slow background Merkle proof anchoring (dual-path pipeline: Security vs Performance) is the primary defense against §103 obviousness.

SECTION V

Filing Status & Next Steps

Filing Status

PATENT	DOCUMENT	STATUS
#1	Original Patent	✓ Approved
#2	Provisional	✓ Drafted
#2	Non-Provisional	✓ Drafted
#3	Provisional	✓ Drafted
#3	Non-Provisional	✓ Drafted
#4	Problem Definition	✓ Complete
#4	Provisional Draft	✓ Drafted
#4	Non-Provisional Draft	✓ Drafted
#5	Problem Definition	✓ Complete
#5	Provisional Draft	✓ Drafted
#5	Non-Provisional Draft	✓ Drafted

Immediate Next Steps

1. **File Patents #2 and #3** — ready now
2. **Insert Five-Layer Trust Stack + Decision Attestation** in Patent #3 non-provisional (secures priority for #4/#5)
3. **Patent #4: Abstraction pass** — broaden independent claim beyond DAG mechanics
4. **Patent #5: Maintain "Trust Health" framing** — strongest asset
5. **Strategic decision confirmed:** Stop after #5, let reality drive #6

Smart Contract

0xA79E149C35Ad47Ed270Bf4b16B80170eBF7B88F8 (Polygon Amoy)